

AlphaTrade India — Master Project Documentation (v1.0)

Author: Product Design & Systems Architecture Draft

Prepared For: AlphaTrade India Project

Domain: Algorithmic Trading Systems (Indian Equity Markets – NSE/BSE)

1. Executive Summary

AlphaTrade India is a professional-grade, automated intraday trading system designed specifically for the Indian stock market. The platform is engineered as a **modular, scalable, and risk-first trading architecture** that enables disciplined algorithmic trading using a small capital base (₹10,000) with intelligent automation, strict capital protection, and real-time monitoring.

The system eliminates emotional trading decisions and human bias by implementing:

- Rule-based execution
- Automated risk controls
- Smart stock selection
- Strategy-driven entries/exits
- Real-time portfolio awareness

The platform follows **institutional-grade design principles** adapted for retail traders.

2. Product Vision

Vision Statement

To build a **fully automated, intelligent intraday trading platform** that operates as a self-governing trading agent for the Indian equity markets, capable of:

- Preserving capital
- Maximizing probability-based returns
- Enforcing discipline
- Operating with minimal human intervention

Core Objectives

- Zero-emotion trading
 - Risk-first architecture
 - Automation by default
 - Modular expansion capability
 - Cloud deployability
 - Professional-grade reliability
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3. System Philosophy

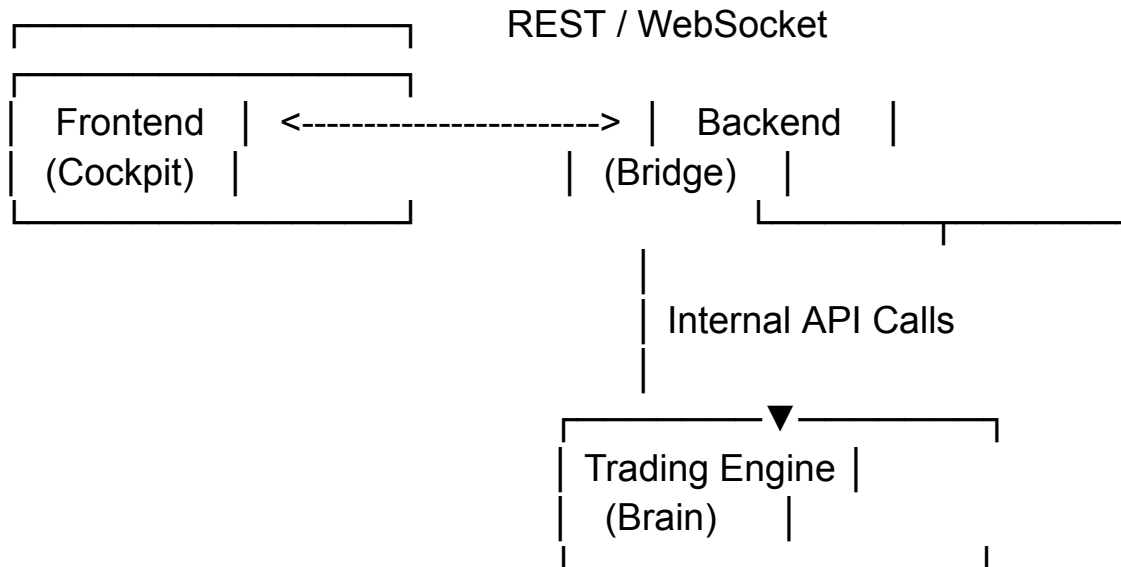
AlphaTrade India follows 4 core principles:

1. **Risk > Profit**
Capital protection is more important than returns.
 2. **Automation > Discretion**
No manual decision making during live trading.
 3. **Data > Emotion**
Only data-driven execution.
 4. **System > Strategy**
System stability matters more than strategy performance.
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4. System Architecture

Architectural Model

Decoupled Three-Tier Architecture



Layer Responsibilities

1. Frontend – "The Cockpit"

Purpose: Human interaction + system control

- Real-time monitoring
- Configuration
- Control triggers
- Emergency actions

2. Backend – "The Bridge"

Purpose: System orchestration

- Authentication
- API routing
- Database management

- Session management
- State control

3. Trading Engine – "The Brain"

Purpose: Autonomous execution

- Market data ingestion
 - Strategy execution
 - Risk control
 - Order management
 - Position management
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5. Technology Stack

Core Stack

Layer	Technology	Purpose
Language	Python 3.10+	Core system language
Backend	FastAPI	Async API server
Frontend	React.js	UI engine
Styling	Tailwind CSS	UI consistency
Database	SQLite + SQLAlchemy	Persistence layer
Broker API	Angel One SmartAPI	Trading execution
Data Stream	WebSockets	Live price feed

Analysis	Pandas, Pandas-TA	Strategy indicators
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6. Core Functional Modules

6.1 Authentication Module

Purpose: Fully automated broker authentication

Features:

- pyotp-based TOTP generation
- Auto-login workflow
- JWT session persistence
- Token refresh scheduler

Flow:

Start System → Generate TOTP → Login → Store Token → Validate Session → Auto Refresh

6.2 Hybrid Stock Selection Engine

Mode 1: Manual Mode

User-selected stocks via UI

- Search-based picker
- Multi-stock selection
- Daily watchlist creation

Mode 2: Intelligence Mode (Auto Scanner)

Scanner Logic:

- Universe: Nifty 100

- Time Window: 09:15–09:20

Filters:

- Open = Low pattern
- High volume breakout
- Price range: ₹100 – ₹1,000
- Liquidity filter

Output:

Ranked trade candidates list

7. Strategy Engine

Strategy Model: ORB + VWAP Fusion

Opening Range Breakout (ORB)

- Range timeframe: First 15 minutes
- Levels:
 - OR High
 - OR Low

VWAP Filter

- Directional confirmation

Entry Conditions

Price > 15-min High

AND

Price > VWAP

Exit Conditions

Type	Logic
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Target +1.5%
Stop -0.75% to
Loss -1%
Time 15:15 PM
Exit

8. Risk Management Architecture

Capital Control

- Base Capital: ₹10,000
- Intraday Leverage: 5x
- Effective Trading Capital: ₹50,000

Risk Rules

Rule	Value
Max Daily Loss	-₹200
Max Trades	User-defined (1–5)
Kill Switch	Immediate square-off
Brokerage Model	₹60 per trade cycle

Risk Engine Logic

If MTM $\leq$ -200 $\rightarrow$ Kill Engine
If Trades $\geq$ Max Trades $\rightarrow$ Lock Trading
If Manual Kill $\rightarrow$ Square Off All Positions

9. Frontend System (React Cockpit)

Core UI Components

Live Metrics Panel

- Real-time P&L
- Margin used
- Available balance
- Active positions

Configuration Panel

- Capital input
- Max trades dropdown
- Strategy selector (future-ready)

Stock Control Panel

- Manual stock picker
- Chips-based UI
- Auto-scan toggle

Emergency Control

- Kill switch (global override)
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10. Data Model Design

Core Tables

trades

- id
- symbol
- entry_price

- exit_price
- qty
- pnl
- timestamp

sessions

- token
- login_time
- expiry

config

- capital
 - max_trades
 - mode
-

11. Folder Architecture

```
alpha-trade-india/
├── backend/
│   ├── main.py
│   ├── auth.py
│   ├── engine.py
│   ├── risk_manager.py
│   ├── models.py
│   └── .env
├── frontend/
│   └── src/
│       ├── components/
│       ├── App.js
│       └── api.js
├── trades.db
└── requirements.txt
```

12. Implementation Roadmap

Phase 1 – Infrastructure

- Repo setup
- API auth
- Broker connection

Phase 2 – Intelligence

- Strategy engine
- Risk manager
- Order execution

Phase 3 – Interface

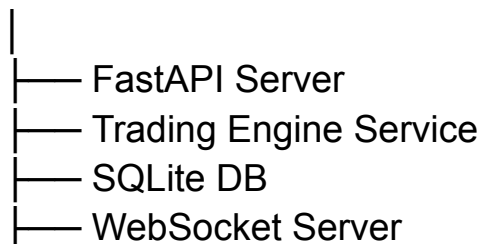
- UI dashboard
- Live metrics
- Control systems

Phase 4 – Production

- Stress testing
 - VPS deployment
 - Capital scaling
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13. Deployment Architecture

Cloud VPS



14. Security & Compliance

- Encrypted API keys
 - Environment variable isolation
 - Rate limiting
 - API throttling
 - Exception handling
 - Auto recovery
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15. Future Expansion Roadmap

- AI-based stock scoring
 - ML prediction engine
 - Reinforcement learning strategy
 - Multi-broker support
 - Portfolio optimization
 - Options trading module
 - Crypto trading module
 - Cloud-native scaling
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16. Product Identity

System Name: AlphaTrade India

Type: Automated Trading Intelligence Platform

Category: Algorithmic Trading System

Market: Indian Equity Markets

User Type: Retail Trader → Professional Trader Transition System

17. Final Statement

AlphaTrade India is not a bot.

It is a **financial automation system**.

It is not a strategy.

It is a **trading intelligence platform**.

It is not a script.

It is a **self-governing trading architecture**.

This system is designed to evolve into a **personal trading AI infrastructure**.